

## Module 8: Convex Geometry — Summary

**What this module is about.** Linear inequalities describe allowed regions rather than single lines. Each inequality selects one half-plane. When several constraints must all be satisfied, the feasible region is their intersection. In the two-variable problems in this module, feasible regions are usually convex polygons. Their vertices are especially important because a linear objective reaches its maximum or minimum at a corner point of a bounded feasible polygon.

A set is **convex** if the complete line segment joining any two points in the set remains inside the set. Filled triangles, rectangles, circles, and half-planes are convex. Crescents, rings, and shapes with inward dents are not.

A linear inequality such as

$$ax + by \leq c$$

has boundary line  $ax + by = c$  and one feasible half-plane. For  $\leq$  or  $\geq$ , the boundary is included. For  $<$  or  $>$ , the boundary is not included.

### Example 1: Find the boundary and test points

Consider

$$x + 2y \leq 8.$$

The boundary line  $x + 2y = 8$  has intercepts

$$(8, 0) \quad \text{and} \quad (0, 4).$$

Test the origin:

$$0 + 2(0) = 0 \leq 8,$$

so the feasible half-plane is the side containing  $(0, 0)$ .

The point  $(2, 3)$  lies on the boundary because  $2 + 2(3) = 8$ . The point  $(4, 3)$  is not feasible because  $4 + 2(3) = 10 > 8$ .

To find a feasible polygon:

1. Graph or understand each boundary line.
2. Determine the feasible side of each inequality.
3. Intersect the allowed half-planes.
4. Find the vertices from axis intercepts and intersections of boundary lines.

A proposed point is feasible only if it satisfies *every* constraint.

**Example 2: Finding a non-axis vertex**

Consider

$$x + y \leq 8, \quad 2x + y \leq 10, \quad x \geq 0, \quad y \geq 0.$$

The two non-axis boundary lines meet where

$$x + y = 8, \quad 2x + y = 10.$$

Subtracting the first equation from the second gives  $x = 2$ , and then  $y = 6$ . One vertex is therefore  $(2, 6)$ . The complete feasible polygon has vertices

$$(0, 0), (5, 0), (2, 6), (0, 8).$$

For a linear objective  $f = ax + by$  on a bounded feasible polygon, at least one maximum and at least one minimum occur at vertices. Therefore, evaluate the objective at every feasible vertex and compare the values.

If two *adjacent* vertices have the same optimal value, then every point on the edge joining them is also optimal.

## Module 8: Convex Geometry — Prequiz

*Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.*

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### A reliable procedure

Boundary line  $\rightarrow$  feasible side  $\rightarrow$  intersection of all constraints  $\rightarrow$  vertices  $\rightarrow$  evaluate the objective.

1. Consider a filled hexagon and a U-shaped region. Which one is convex? Use the definition of convexity to explain your answer in one sentence.

2. A workshop constraint is

$$2x + y \leq 10,$$

where  $x$  and  $y$  are nonnegative quantities.

- (a) Find the positive  $x$ - and  $y$ -intercepts of the boundary line.
- (b) Determine whether  $(3, 4)$  is feasible.
- (c) Determine whether  $(5, 3)$  is feasible.

3. Consider the system

$$x + y \leq 9, \quad 2x + y \leq 12, \quad x \geq 0, \quad y \geq 0.$$

Find the intersection of the two non-axis boundary lines. Then list all vertices of the feasible region.

4. Using the feasible vertices from Problem 3,

$$(0, 0), (6, 0), (3, 6), (0, 9),$$

evaluate

$$P = 5x + 4y$$

at every vertex. State the maximum value and the vertex where it occurs.

5. A feasible polygon has vertices  $(0, 0)$ ,  $(5, 0)$ ,  $(3, 3)$ ,  $(0, 6)$ . Evaluate  $f = x + y$  at all four vertices. Two adjacent vertices tie for the maximum. What does that imply about the edge connecting them?

## Module 8: Convex Geometry — Matching Quiz

*These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.*

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1. Which is convex: a filled triangle or a ring-shaped region? Explain using line segments between points in the set.

2. For the constraint  $3x + y \leq 12$ , find the positive intercepts and determine whether  $(3, 4)$  is feasible.

3. Find all vertices of the feasible region

$$x + y \leq 7, \quad 2x + y \leq 10, \quad x \geq 0, \quad y \geq 0.$$

Show the calculation for the intersection of the two slanted boundary lines.

4. Evaluate  $R = 5x + 4y$  at the vertices  $(0, 0)$ ,  $(5, 0)$ ,  $(3, 4)$ ,  $(0, 7)$ . Find the maximum and the vertex where it occurs.

5. Suppose a linear objective has the same maximum value at two adjacent vertices of a feasible polygon. Describe all of the optimal points.